

Will Ruff

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EDUCATION

San Diego State University

Bachelor of Science in Computer Science, Minor in Mathematics

Major GPA: 4.0 | Cumulative GPA: 3.88

San Diego, CA

Aug. 2022 – May 2026

RELEVANT COURSEWORK

- Machine Learning
- Algorithms
- Object Oriented Programming
- Data Science
- Adv. Programming Languages
- Database Theory Implementation
- Intro to Artificial Intelligence
- Operating Systems
- Data Structures

EXPERIENCE

Data Engineer Intern

Kenworth - Division of PACCAR Inc.

May 2025 – Aug. 2025

Seattle, WA

- Engineered an in-house computer vision system replacing a third-party tool, reducing costs by \$100K+ annually
- Increased cab part detection accuracy from 94.5% → 96.5% through iterative training and hyperparameter tuning
- Developed custom YOLO and Azure CV models for part detection and cab positioning along the production line
- Built a Flask web application to deliver real-time inference to production dashboards, aiding operator inspections
- Designed a YOLO camera-triggering model to detect optimal cab alignment for capturing all four sides at EOL
- Created a real-time part detection stream with live bounding boxes on cabs as they exit the manufacturing line

Operating Systems Teaching Assistant

San Diego State University

Jan. 2025 – May 2025

San Diego, CA

- Assisted in maintaining and configuring the auto-grader codebase to ensure accurate and efficient grading
- Set up and tested the auto grader for each assignment, optimizing performance and creating unique test cases
- Graded assignments and exams for a class of 150 students ensuring consistency and adherence to grading rubrics
- Held weekly office hours to provide personalized support, debugging assistance, and address student questions
- Collaborated with the instructor to improve course materials and enhance the students' learning experience

Software Engineer Intern

Glympse - Location Sharing Technology

June 2024 – Aug. 2024

Seattle, WA

- Developed Python and Go scripts for file comparison, cutting memory usage by 25% via benchmarking
- Migrated legacy code to a new repository, adding essential parameters and enhancing report functionality
- Refactored and modernized a shell script report in Python, enhancing readability and maintainability
- Built and implemented comprehensive unit tests to ensure the reliability and accuracy of the Python code
- Deployed code using GitLab CI/CD pipelines, configuring environments with '.gitlab-ci.yml' and Docker

PROJECTS

Fitness Pose Detection App | *Core ML, Swift, Python*

Jul. 2025 – Aug. 2025

- Engineered a custom object detection model with Core ML to classify 8 bodybuilding poses with high accuracy
- Integrated Core ML into an iOS app for on-device inference and real-time pose recognition during photo capture
- Automated web scraping to collect a large dataset of bodybuilding pose photos for annotation and model training
- Built a scalable photo labeling pipeline, tagging and storing poses with user weight and date metadata for tracking
- Optimized model performance for mobile, balancing accuracy and speed for real-time use and bulk imports

StockScript | *Python, TextX*

Nov. 2024 – Dec. 2024

- Designed and implemented a domain-specific programming language for stock and crypto modeling
- Developed grammar definitions using TextX and created a Python interpreter to execute scripts effectively
- Created user-friendly stock and crypto declarations with operations like price changes, merges, and splits
- Developed display features to present asset data, including prices, market caps, and supply metrics
- Added a simulation mode to generate good, bad, or random days for specific stocks or the entire market

TECHNICAL SKILLS

Languages: Python, C++, Java, JavaScript, HTML/CSS, TextX

Frameworks & Technologies: Git, YOLO, Core ML, Azure Custom Vision, Object Detection, Docker, Flask, CI/CD Pipelines, OpenCV, Tensorflow, Jupyter, Linux, VirtualBox, VS Code, GNU Debugger, unittest, Azure Data Factory